

Ques. HOW IS LED DRIVER DIFFERENT FROM SMPS AND IGBT FROM MOSFET? HOW TO TEST AN IGBT USING MULTIMETER?

Abhijit Dasgupta

Ans. LEDs work most efficiently and safest with a constant-current drive. As a result, many new devices have been developed to provide this type of LED drive. LED power sources that provide a constant-current output have typically been referred to as LED drivers.

An LED driver responds to the changing needs of the LED, or LED circuit, by providing a constant quantity of power to the LED as its electrical properties change with temperature. An LED driver is a self-contained power supply, which has outputs that are matched to the electrical characteristics of the LED or LEDs.

LED drivers may offer dimming by means of pulse width modulation circuits and may have more than one channel for separate control of different LEDs or LED arrays. The power level of the LED is maintained constant by the LED driver as the electrical properties change throughout the temperature increases and decreases seen by the LED or LEDs. Without the proper driver, the LED may become too hot and unstable, causing poor performance or failure.

A conventional SMPS provides an output that is regulated to give a constant voltage. If your LED applications are just for experimenting, you may use an SMPS and control the current by limiting resistors in series to the LEDs.

You may refer to the following links for details:

<http://power-topics.blogspot.in/2011/08/what-type-of-led-driver-or-power-supply.html>

<http://www.edaboard.com/thread157546.html>

Differences between IGBT and MOSFET are many. Some of these are:

1. Although both IGBT and MOSFET are voltage-controlled devices, IGBT has BJT-like conduction characteristics.

2. Terminals of IGBT are known as emitter, collector and gate, whereas MOSFET has gate, source and drain.

3. IGBTs are better in power handling than MOSFETs.

4. IGBT has PN junctions; MOSFET does not have these. It is a physical characteristic which affects the working function.

5. IGBT has a lower forward voltage drop compared to MOSFET.

6. MOSFETs have higher switching frequencies and hence these are preferred over IGBTs in power supplies like SMPS.

For details please refer to the following links:

<http://www.differencebetween.com/difference-between-igt-and-vs-mosfet/>

https://en.wikipedia.org/wiki/MOSFET#/media/File:MOSFET_functioning_body.svg

<http://electronicdesign.com/power/what-s-difference-between-igbt-and-high-voltage-power-mosfets>

To test an IGBT with a multimeter, the meter should be used in diode-checking mode. It should have battery voltage of less than 20V (a typical unit using 9V battery is alright).

1. To test collector-emitter junction:

1. With the module out of circuit, remove the conductive foam and short the gate to emitter.

2. With multimeter in diode-check mode, the collector to emitter should give a normal diode reading with positive on the emitter and negative on the collector.

3. The multimeter should read open or infinite with positive on the collector and negative on the emitter. Damaged IGBTs may test as shorted in both positive and negative directions, open in both directions, or resistive in both directions.

For gate oxide test: With multimeter in resistance mode, the resistance from gate to collector and gate to emitter should be infinite on a good device. A damaged device may appear shorted or have resistive leakage from gate to collector and/or emitter.

For details please check:

https://www.galco.com/circuit/igt_testing.htm

Q2. WHY DO WE USE A COMMON GROUND TRACK FOR ALL THE PERIPHERAL CONNECTORS OF A PC?

A. Samiuddin

A2. All ground connections of a PC's peripherals are made directly to the ground plane of an SMPS PCB to prevent ground loops. Loops can produce large amounts of radiated emission, which is unsuitable for electromagnetic compatibility. So we use a common ground track for all peripheral connectors instead of using more than one ground wire.

For more information on the subject please check:

http://www.onsemi.com/pub_link/Collateral/AND8301-D.PDF

Answers compiled by EFY senior application engineer, Nidhi Kathuria. Letters and questions for publication may be addressed to Editor, Electronics For You, D-97/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editor@efy.in) and should include name and address of the sender